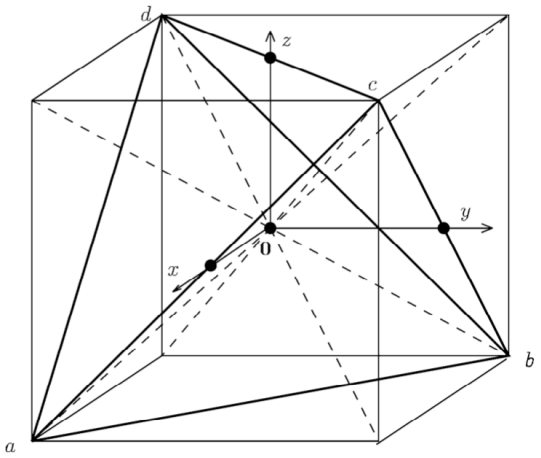
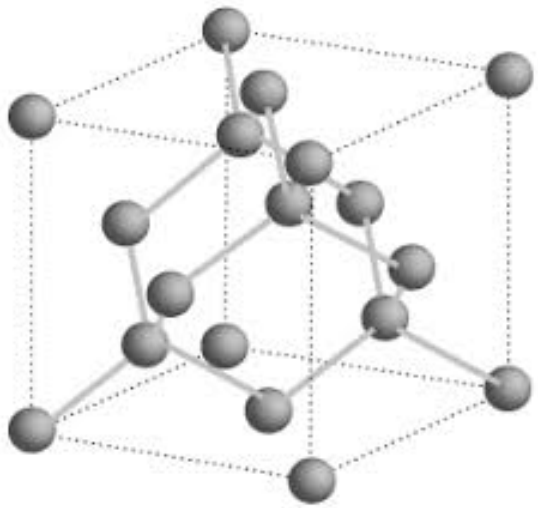
Spin orbit coupling in solids

Effect of spin: Group theory

- Example: Ge ($O_h \cong O \otimes i \cong T_d \otimes i$)



T_d ($\bar{4}3m$)		E	$8C_3$	$3C_2$	$6\sigma_d$	$6S_4$
$x^2+y^2+z^2$	A_1	1	1	1	1	1
	A_2	1	1	1	-1	-1
$(x^2-y^2, 3z^2-r^2)$	E	2	-1	2	0	0
(R_x, R_y, R_z) $yz, zx, xy)$	T_1	3	0	-1	-1	1
(x, y, z)	T_2	3	0	-1	1	-1



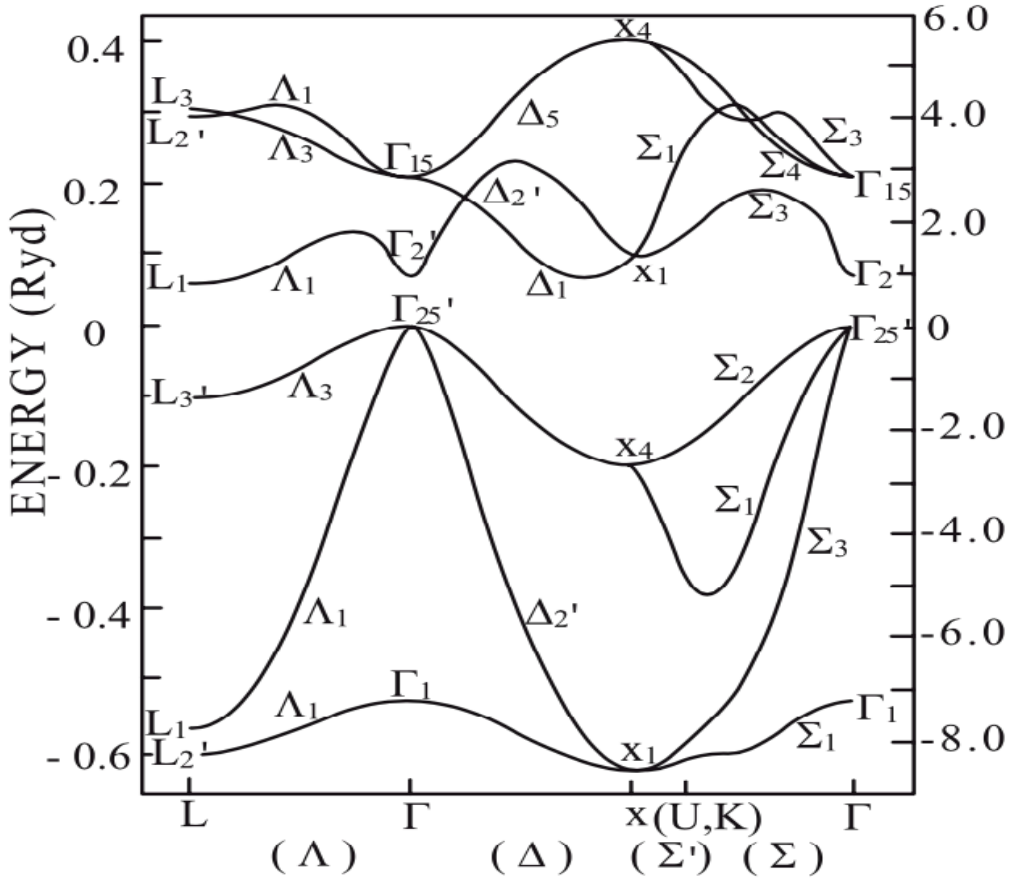
Effect of spin: Group theory

- Ge non-symmorphic space group

representation	$\{E\|0\}$	$3\{C_4^2\|0\}$	$6\{C_4\|\tau_d\}$	$6\{C_{2'}\|\tau_d\}$	$8\{C_3\|0\}$	$\{i\|\tau_d\}$	$3\{iC_4^2\|\tau_d\}$	$6\{iC_4\|0\}$	$6\{iC_{2'}\|0\}$	$8\{iC_3\|\tau_d\}$
Γ_1	1	1	1	1	1	1	1	1	1	1
Γ_2	1	1	-1	-1	1	1	1	-1	-1	1
Γ_{12}	2	2	0	0	-1	2	2	0	0	-1
Γ_{15}	3	-1	1	-1	0	-3	1	-1	1	0
Γ_{25}	3	-1	-1	1	0	-3	1	1	-1	0
Γ_1'	1	1	1	1	1	-1	-1	-1	-1	-1
Γ_2'	1	1	-1	-1	1	-1	-1	1	1	-1
Γ_{12}'	2	2	0	0	-1	-2	-2	0	0	1
Γ_{15}'	3	-1	1	-1	0	3	-1	1	-1	0
Γ_{25}'	3	-1	-1	1	0	3	-1	-1	1	0

Effect of spin: Group theory

- Ge band structure without spin



Effect of spin: Group theory

$$\chi_j(\alpha) = \frac{\sin(j+1/2)\alpha}{\sin(\alpha/2)} \qquad \chi_j(\alpha+2\pi) = \chi_j(\alpha)(-1)^{2j}$$

Half interger spin

$$\chi_j(\alpha \pm 2\pi) = -\chi_j(\alpha)$$

$$\chi_j(\alpha \pm 4\pi) = +\chi_j(\alpha)$$

Intrinsic degree of freedom

Intrinsic 2 pi rotation R

Effect of spin: Group theory

- Ge Spin double group (one quarter)

O	E	$\mathcal{R}$	$3C_4^2 + 3\mathcal{R}C_4^2$	$6C_4$	$6\mathcal{R}C_4$	$6C_2' + 6\mathcal{R}C_2''$	$8C_3$	$8\mathcal{R}C_3$
Γ_1	1	1	1	1	1	1	1	1
Γ_2	1	1	1	-1	-1	-1	1	1
Γ_{12}	2	2	2	0	0	0	-1	-1
$\Gamma_{15'}$	3	3	-1	1	1	-1	0	0
$\Gamma_{25'}$	3	3	-1	-1	-1	1	0	0
Γ_6	2	-2	0	$\sqrt{2}$	$-\sqrt{2}$	0	1	-1
Γ_7	2	-2	0	$-\sqrt{2}$	$\sqrt{2}$	0	1	-1
Γ_8	4	-4	0	0	0	0	-1	1

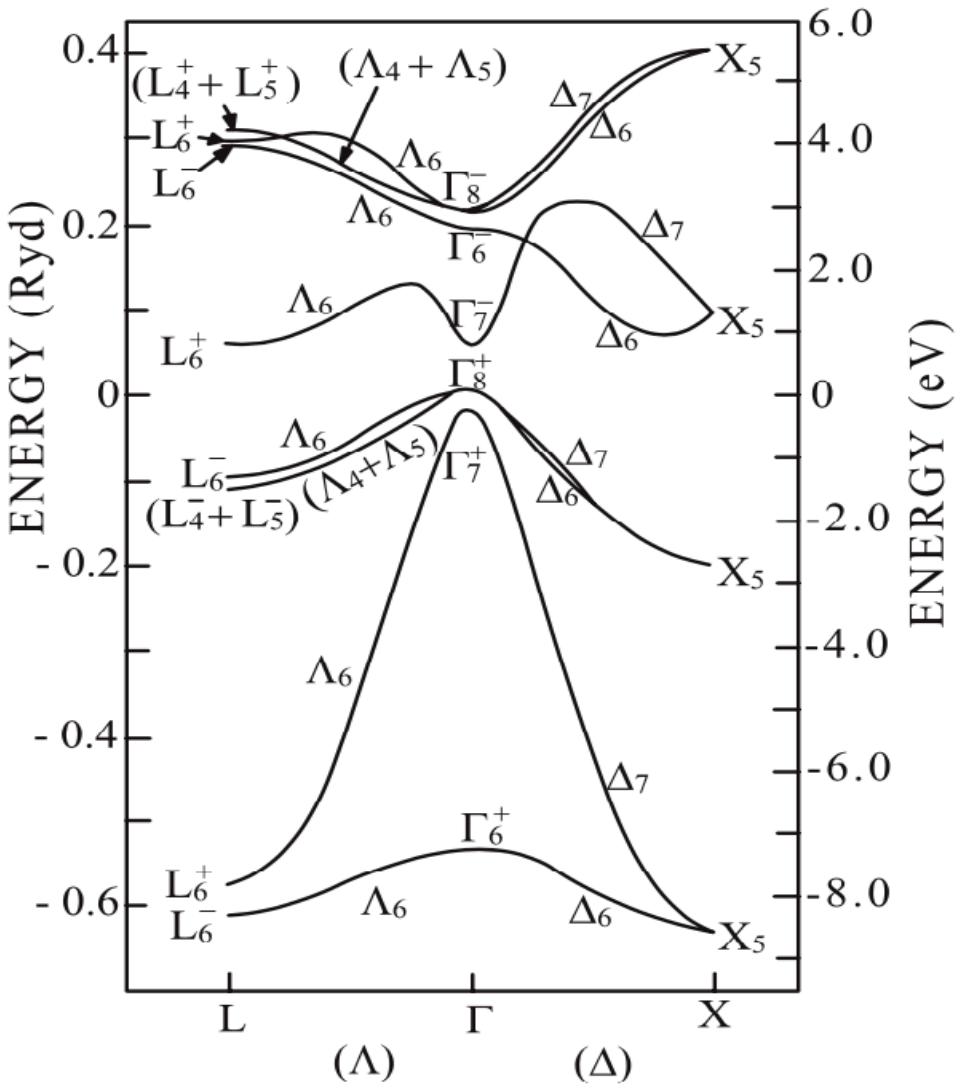
Effect of spin: Group theory

- $\chi_{\frac{1}{2}}(\alpha) = 2\cos\left(\frac{\alpha}{2}\right),\ \chi_{\frac{1}{2}}(0) = 2, \chi_{\frac{1}{2}}(\pi) = 0, \chi_{\frac{1}{2}}\left(\frac{\pi}{2}\right) = \sqrt{2}, \chi_{\frac{1}{2}}\left(\frac{\pi}{3}\right) = 1$
- Spacial part x spin part $\rightarrow\ \Gamma_i \otimes \Gamma_6^+$
- Band spliting

$\Gamma_1^+ \otimes \Gamma_6^+ = \Gamma_6^+$	$\Gamma_1^- \otimes \Gamma_6^+ = \Gamma_6^-$
$\Gamma_2^+ \otimes \Gamma_6^+ = \Gamma_7^+$	$\Gamma_2^- \otimes \Gamma_6^+ = \Gamma_7^-$
$\Gamma_{12}^+ \otimes \Gamma_6^+ = \Gamma_8^+$	$\Gamma_{12}^- \otimes \Gamma_6^+ = \Gamma_8^-$
$\Gamma_{15}^+ \otimes \Gamma_6^+ = \Gamma_6^+ + \Gamma_8^+$	$\Gamma_{15}^- \otimes \Gamma_6^+ = \Gamma_6^- + \Gamma_8^-$
$\Gamma_{25}^+ \otimes \Gamma_6^+ = \Gamma_7^+ + \Gamma_8^+$	$\Gamma_{25}^- \otimes \Gamma_6^+ = \Gamma_7^- + \Gamma_8^-$
$\Gamma_6^+ \otimes \Gamma_6^+ = \Gamma_1^+ + \Gamma_{15}^+$	$\Gamma_6^- \otimes \Gamma_6^+ = \Gamma_1^- + \Gamma_{15}^-$
$\Gamma_7^+ \otimes \Gamma_6^+ = \Gamma_2^+ + \Gamma_{25}^+$	$\Gamma_7^- \otimes \Gamma_6^+ = \Gamma_2^- + \Gamma_{25}^-$
$\Gamma_8^+ \otimes \Gamma_6^+ = \Gamma_{12}^+ + \Gamma_{15}^+ + \Gamma_{25}^+$	$\Gamma_8^- \otimes \Gamma_6^+ = \Gamma_{12}^- + \Gamma_{15}^- + \Gamma_{25}^-$

Effect of spin: Group theory

- Ge band structure (with spin)
- $\Gamma_{15}^{-} \otimes \Gamma_{6}^{+}(6) = \Gamma_{6}^{-}(2) + \Gamma_{8}^{-}(4)$
- $\Gamma_{25}^{+} \otimes \Gamma_{6}^{+}(6) = \Gamma_{7}^{+}(2) + \Gamma_{8}^{+}(4)$



Time reversal symmetry(T) & Inversion symmetry(P)

T

$k \to -k\,, \qquad \sigma \to -\sigma$

$$E_{n\uparrow}(\boldsymbol{k}) = E_{n\downarrow}(-\boldsymbol{k})$$

$$E_{n\downarrow}(\boldsymbol{k}) = E_{n\uparrow}(-\boldsymbol{k})$$

P

$k \to -k\,, \qquad \sigma \to \sigma$

$$E_n(\boldsymbol{k}) = E_n(-\boldsymbol{k})$$

T, *P*

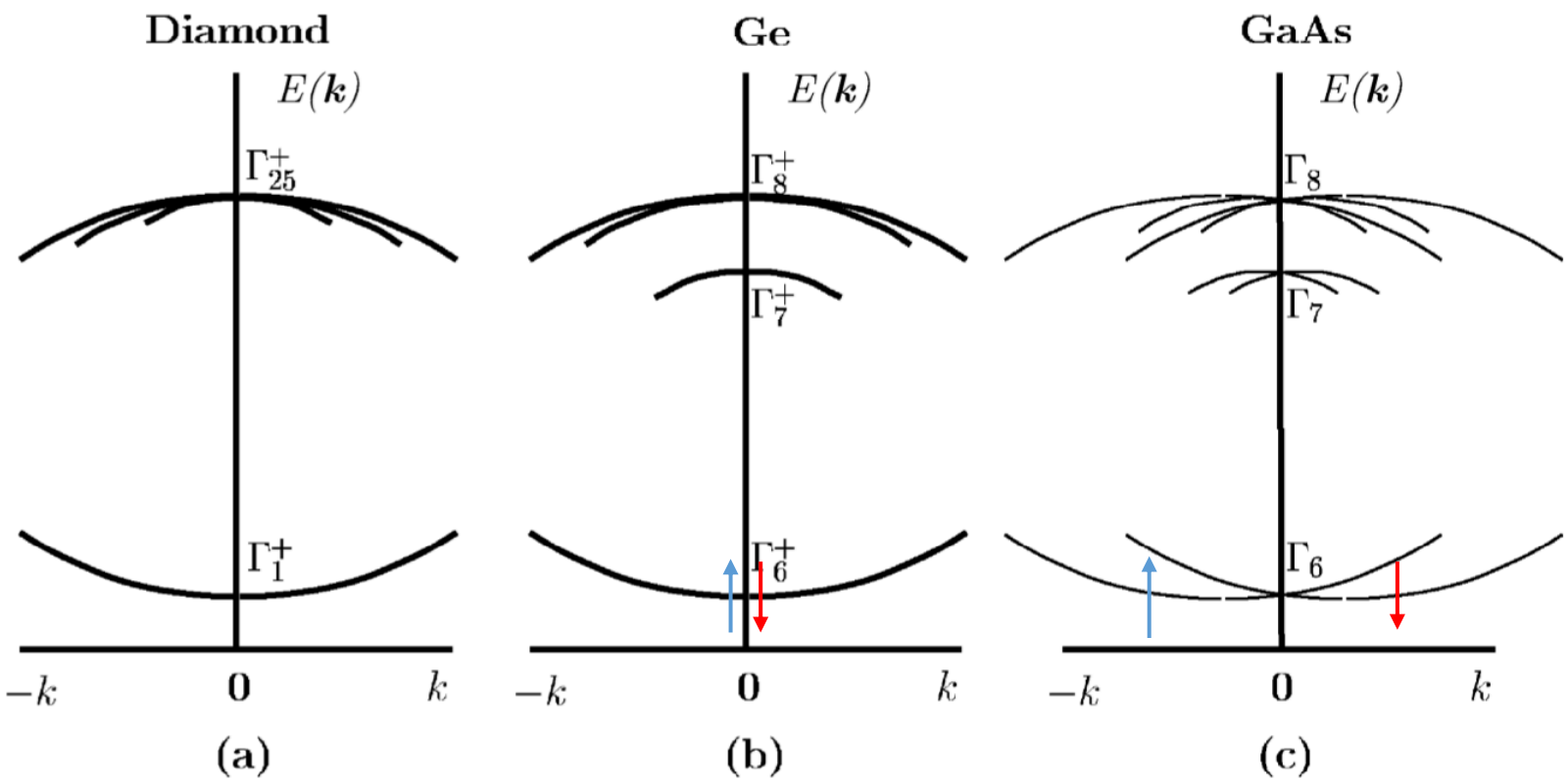
$$E_{n\uparrow}(\boldsymbol{k}) = E_{n\downarrow}(-\boldsymbol{k}) \;=\; E_{n\downarrow}(\boldsymbol{k}) = E_{n\uparrow}(-\boldsymbol{k})$$

T, no *P*

$$E_n(\boldsymbol{k}) \;\neq\; E_n(-\boldsymbol{k})$$

Time reversal symmetry & Inversion symmetry

no spin with spin,T,P with spin,T,no P



Dresselhauls term (BIA)

- Dresselhauls(Bulk inversion asymmetry(BIA))
- Zinc-blende

$$H_D^{3\mathrm{D}} \propto \left[p_x(p_y^2 - p_z^2)\sigma_x + p_y(p_z^2 - p_x^2)\sigma_y + p_z(p_x^2 - p_y^2)\sigma_z\right]$$

- Surface effective Hamiltonian

$$H_{D,\mathrm{linear}}^{2\mathrm{D},001} = \beta\left[-p_x\sigma_x + p_y\sigma_y\right]$$

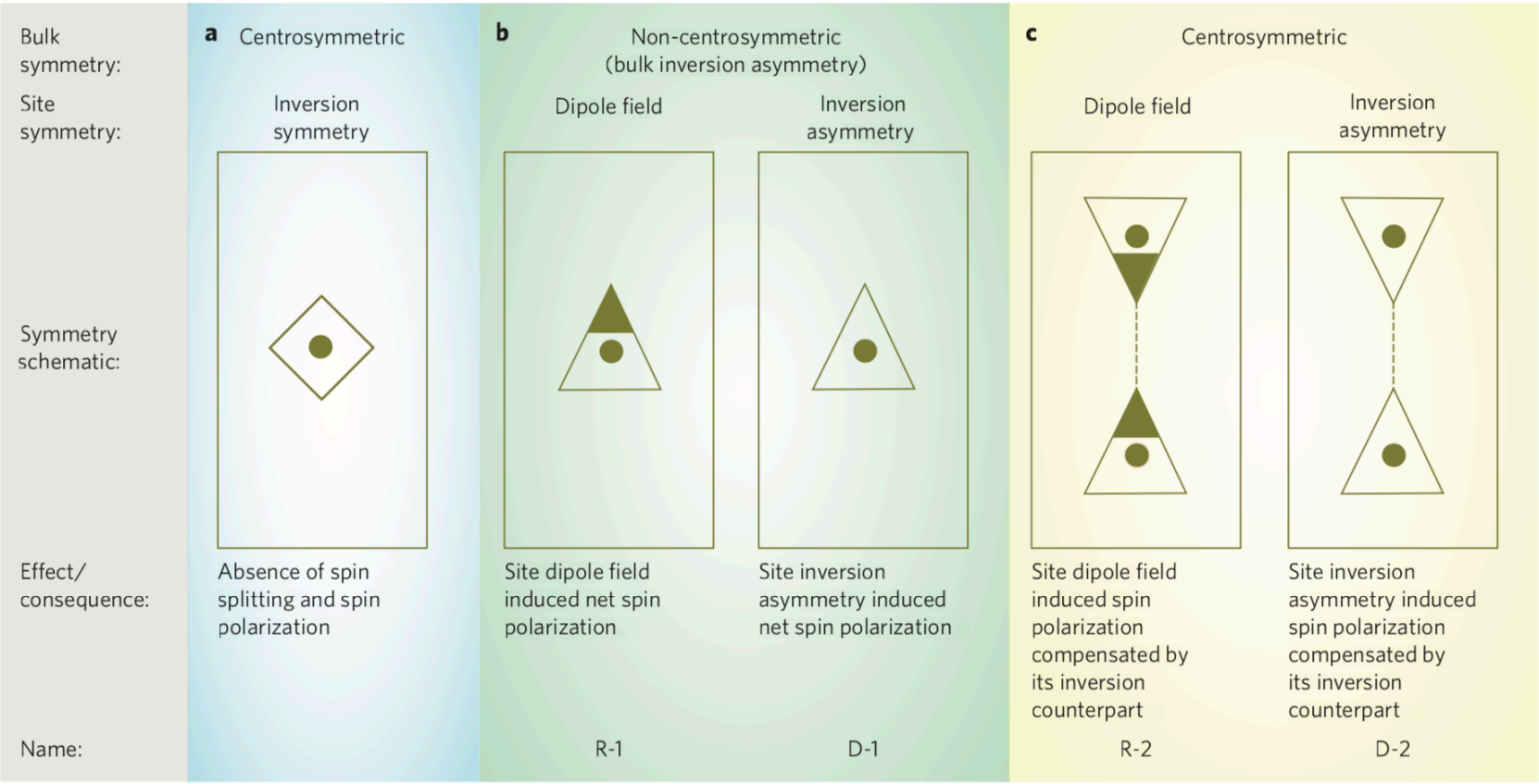
Rashba term (SIA)

- Usually in heterstructure surface

$$H_R \propto [\vec{E} \times \vec{p}] \cdot \vec{\sigma} = E_z(-p_y\sigma_x + p_x\sigma_y) = \alpha(-p_y\sigma_x + p_x\sigma_y)$$

Hidden spin polarization

- *Hidden spin polarization in inversion-symmetric bulk crystals* Xiuwen Zhang, Qihang Liu, Junwei Luo, Arthur J. Freeman& Alex Zunger *Nature Physics* **volume10**, pages387–393 (2014)



Classification of spin polarization

Table 1 | Classification of spin polarization in nonmagnetic bulk materials on the basis of bulk space group and site point group.

Site point group / Bulk space group	Non-centrosymmetric (at least one site)			Centrosymmetric (all sites)
	Non-polar (all sites) (D_2, D_3, D_4, D_6, S_4, D_{2d}, C_{3h}, D_{3h}, T, T_d, O)	Polar (at least one site) (C_1, C_2, C_3, C_4, C_6, C_{1V}, C_{2V}, C_{3V}, C_{4V}, C_{6V})		(C_i, C_{2h}, D_{2h}, C_{4h}, D_{4h}, S_6, D_{3d}, C_{6h}, D_{6h}, T_h, O_h)
		Dipoles add up to zero	Dipoles add up to non-zero	
Non-centrosymmetric (for example, $F\bar{4}3m$)	**a** D-1 Example: GaAs, ZrCoBi	**b** D-1 Example: γ-$LiAlO_2$	**c** R-1 & D-1 Example: BiTeI, α-SnTe	*Not possible* (*Site point group* cannot be centrosymmetric *if space group is non*-centrosymmetric)
Centrosymmetric (for example, $R\bar{3}m$)	**d** D-2 Example: Si, NaCaBi	**e** R-2 & D-2 Example: MoS_2, Bi_2Se_3, $LaOBiS_2$		**f** Absence of spin polarization Example: β-SnTe